

PROFESSIONAL SUMMARY

Systems Engineering undergraduate experienced in algorithmic design, object-oriented software development, and embedded control systems. Focused on building scalable logic, ensuring system reliability, and delivering seamless integration between software and hardware environments.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, MATLAB, Julia, HTML/CSS/JavaScript, CLINGO.
- **Hardware & Embedded:** Microcontroller Programming (Arduino Uno R3, Cypress PSoC), Master-Slave Architectures, Analog & Digital Electronics, Hardware Communication Protocols (SPI, I2C, Bluetooth), Sensor & Actuator Integration (MQ-7, PIR, DHT11, Moisture, Servos, H-Bridge / DC Motors).
- **Systems & Automation:** Control Theory & Design (PID, Bode/Nyquist, Routh-Hurwitz), State-Space Modeling, System Identification, Signals & Systems (DSP, FFT, Laplace/Z-Transforms), MATLAB/Simulink Modeling.
- **Algorithms & Software Design:** OOP, Data Structures (Trees: AVL/BST, Hash Tables, Heaps), Graph Algorithms (DFS/BFS), Algorithmic Paradigms (Dynamic Programming, Two Pointers), Multithreading, Non-blocking concepts.
- **Tools & Environment:** Git, GitHub, VS Code, Linux/Bash, Node.js (npm), Arduino IDE, PSoC Creator, MATLAB Control/Robust Toolboxes, AutoCAD, Netica.

ENGINEERING & SOFTWARE PROJECTS

Automated Smart Home Safety & Exhaust System | C++, Arduino, OOP

- Engineered an automated safety system using an Arduino Uno to monitor hazardous gas levels (MQ-7) and automatically trigger a DC ventilation fan and a servo-driven window mechanism.
- Structured the C++ firmware into modular OOP classes, utilizing a time-based millis() state machine to enable concurrent sensor reading and smooth, incremental servo actuation.
- Implemented an autonomous recovery cycle that continuously evaluates air quality, automatically resetting actuators (closing the window and stopping the fan) once the gas concentration drops below a safely calibrated threshold.

Distributed Smart Irrigation System | Embedded C++, SPI, Bluetooth, Hardware Interrupts

- Developed a distributed IoT automation system utilizing two Arduino microcontrollers communicating via the Master/Slave SPI protocol, allowing an Arduino (Master) processing environmental data (Bluetooth, DHT11) to control a second Arduino (Slave) managing irrigation logic.
- Programmed hardware timers (Timer1) and Interrupt Service Routines (ISRs) to instantly override actuators based on real-time PIR motion detection, ensuring fail-safe operation.
- Structured the embedded codebase strictly using Object-Oriented C++ principles, encapsulating hardware states into reusable, modular classes.

Algorithmic Data Pipeline ("The Royal Hunt") | C, Algorithms & Data Structures

- Architected a multi-stage data processing pipeline in C, intelligently chaining Queues and Binary Search Trees (BST) to parse, filter, and sort raw data extracted from CSV files.
- Optimized data retrieval and dynamic ranking by implementing Hash Tables for O(1) record lookups and Max Heaps to efficiently update attributes and extract top-performing entities.
- Implemented Depth-First Search (DFS) graph traversal algorithms to calculate and evaluate all valid network routes for automated entity assignment based on dynamic criteria.

Enterprise Cloud Calendar System | Java, OOP, Data Structures

- Developed a cloud-based calendar scheduling application in Java (similar to Google Calendar), implementing a robust object-oriented architecture to manage overlapping events, fixed-duration meetings, and participant responses.
- Implemented a multi-tier account system (Staff, Managers, Guests) that enforces specific business rules—such as priority-based scheduling limits, automatic high-priority invitation acceptance, and conflict overrides.
- Designed a modular codebase using object-oriented principles to handle event identification, multi-user invitations, and real-time status tracking (accepted, declined, pending).

AI & Predictive Modeling Models | Netica, Clingo, Algorithmic Search

- Implemented optimization & search algorithms (Simulated Annealing, A*, MinMax), Logic Programming, and Bayesian Networks for predictive modeling.

EDUCATION

Faculty of Automatic Control and Computers , University POLITEHNICA of Bucharest | *Expected 2028*

B.Sc. Systems Engineering (*Automation and applied informatics*)

+ *Pedagogical Training Module (Ongoing)*

Erasmus+ Mobility Program | *NOVA School of Science and Technology*

Colegiul Național „Mihai Viteazul”, București

- Baccalaureate: Mathematics (10), Informatics (9.95), Romanian (9.65).